

Problem 6.17

Joe deposits 10 today and another 30 in 5 years into a fund that pays *simple interest* at 11% per year. Tina will make the same two deposits, but the 10 will be deposited n years from today and the 30 will be deposited $2n$ years from today. Tina's deposits earn an annual effective rate of 9.15%. At the end of 10 years, the accumulated amount of Tina's deposits is equal to the accumulated amount of Joe's deposits. Calculate n .

Solution

First, we accumulate Joe's deposits to the end of year 10.

Since Joe's fund pays simple interest at 11% per year,

$$10(1 + 0.11 \cdot 10) = 21$$

and

$$30(1 + 0.11 \cdot 5) = 46.5.$$

Therefore, Joe's total accumulated value at the end of 10 years is

$$21 + 46.5 = 67.5.$$

Now we accumulate Tina's deposits to the end of year 10. Since her rate is an annual effective rate of 9.15%, we have

$$10(1.0915)^{10-n} + 30(1.0915)^{10-2n} = 67.5.$$

Dividing both sides by $(1.0915)^{10}$,

$$10(1.0915)^{-n} + 30(1.0915)^{-2n} = \frac{67.5}{(1.0915)^{10}} \approx 28.13.$$

Make the substitution

$$x = (1.0915)^{-n}.$$

Then the equation becomes

$$10x + 30x^2 = 28.13,$$

or equivalently,

$$30x^2 + 10x - 28.13 = 0.$$

Solving the quadratic equation,

$$x \approx 0.8159$$

(the other root is negative and is not valid in this context).

Thus,

$$(1.0915)^{-n} = 0.8159.$$

Taking logarithms,

$$\ln(0.8159) = -n \ln(1.0915).$$

Therefore,

$$n = -\frac{\ln(0.8159)}{\ln(1.0915)} \approx 2.32.$$

$$\boxed{n \approx 2.32 \text{ years}}$$